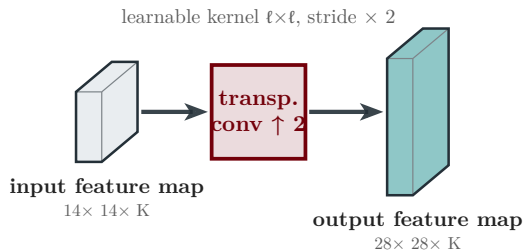


Transposed convolution — learnable upsampling

spatial size grows $\times 2$ ($s = 2$)



How one input cell upsamples: multiply the shared kernel, stamp into the output

input 2×2 kernel $\ell \times \ell$ output 3×3

a	b
c	d

$\times$

w_0	w_1
w_2	w_3

stamp & add

a w_0	a w_1	
a w_2	a w_3	

other cells stamp at stride-shifted spots; overlaps summed